§ Delta Method

Motivation

Fundamental theorem of Calculus

$$f(x) - f(a) = \int_a^x f'(t)\,dt \left(= \int_a^x f'(t)\,d(t-x) \right)$$

$$= f'(t)(t-x)\Big|_a^x - \frac{1}{2}\int_a^x f''(t)\,d(t-x)^2$$

$$= f'(a)(x-a) - \frac{1}{2}\left\{ f''(t)(t-x)^2\Big|_a^x - \frac{1}{3}\int_a^x f^{(3)}(t)\,d(t-x)^3 \right\}$$

$$\vdots$$

$$= \sum_{k=1}^{n} \frac{f^{(n)}(a)}{n!}(x-a)^n + (-1)^n \int_a^x \frac{f^{(n+1)}(a)}{n!}(t-x)^n\,dt$$

$$\Rightarrow f(x) = \sum_{k=0}^{n} \frac{f^{(n)}(a)}{n!}(x-a)^n + \int_a^x \frac{f^{(n+1)}(a)}{n!}(x-t)^n\,dt$$

$$= T_n(x) + R_n(x)$$

Reminder associated w/ $T_n(x)$

Taylor polynomial of order n about a.

Generalized to multivariate

Gradient of f evaluated at $x = a$

Hessian matrix

$$f(x) = f(a) + \nabla f(a)^T (x-a) + (x-a)^T H_f(a)(x-a) + o(\|x-a\|^2)$$

little o : $f(x) = o(g(x))$ as $x \to \infty$

$$\Leftrightarrow \lim_{x\to\infty} \frac{f(x)}{g(x)} = 0$$

For 1st order Taylor series of g about θ ($E(X) = \theta$)

$$g(x) \approx g(\theta) + \nabla g(\theta)^T (x-\theta)$$

then

$$E[g(x)] \approx g(\theta)$$

$$V[g(x)] \approx \nabla g(\theta)^T V(x)\, \nabla g(\theta)$$

Ex. Mediation Model

$$M = \alpha_0 + \alpha_1 A + \varepsilon_M$$
$$Y = \theta_0 + \theta_1 M + \theta_2 A + \varepsilon_Y$$

the indirect effect is the form

$$IE = \alpha_1 \theta_1$$

Let $g(a,b) = ab$ and $\underset{\sim}{\theta} = \begin{bmatrix} \alpha_1 \\ \theta_1 \end{bmatrix} \Rightarrow g(\underset{\sim}{\theta}) = \alpha_1 \theta_1$

Expand g around true $\underset{\sim}{\theta}$

$$g(\hat{\underset{\sim}{\theta}}) \approx g(\underset{\sim}{\theta}) + \nabla g(\underset{\sim}{\theta})^T (\hat{\underset{\sim}{\theta}} - \underset{\sim}{\theta})$$

which means

$$\mathbb{E}(\hat{\alpha}_1 \hat{\theta}_1) \approx \alpha_1 \theta_1$$

and

$$\nabla g(\underset{\sim}{\theta}) = \begin{bmatrix} \theta_1 \\ \alpha_1 \end{bmatrix}$$

$$\nabla g(\underset{\sim}{\theta})^T V(\hat{\underset{\sim}{\theta}}) \nabla g(\underset{\sim}{\theta}) = [\theta_1 \ \ \alpha_1] \begin{bmatrix} V(\hat{\alpha}_1) & Cov(\hat{\alpha}_1, \hat{\theta}_1) \\ Cov(\hat{\alpha}_1, \hat{\theta}_1) & V(\hat{\theta}_1) \end{bmatrix} \begin{bmatrix} \theta_1 \\ \alpha_1 \end{bmatrix}$$

$$= \theta_1^2 V(\hat{\alpha}_1) + \alpha_1^2 V(\hat{\theta}_1) + 2\alpha_1\theta_1 Cov(\hat{\alpha}_1, \hat{\theta}_1)$$

$$= \theta_1^2 V(\hat{\alpha}_1) + \alpha_1^2 V(\hat{\theta}_1)$$

? $\varepsilon_M \perp \varepsilon_Y$

$\Rightarrow Cov(\hat{\alpha}_1, \hat{\theta}_1) = 0$

Under the null

$$H_0 : \alpha_1 \theta_1 = 0$$

the Sobel test statistic

$$\frac{\hat{\alpha}_1 \hat{\theta}_1}{\left[\hat{\alpha}_1^2 V(\hat{\theta}_1) + \hat{\theta}_1^2 V(\hat{\alpha}_1)\right]^{\frac{1}{2}}} \sim N(0,1)$$

Subject : ....

§ Hypothesis Testing

If θ denotes a population parameter, the general format of null hypothesis and alternative hypothesis is

$$H_0 : \theta \in \Theta \quad \text{vs} \quad H_1 : \theta \in \underbrace{\Theta^c}_{\text{complement}}$$

UIT
$\Updownarrow$

Def (IUT : Intersection-Union Test)

Suppose H_0 is expressed as a union.

$$H_0 : \theta \in \bigcup_{d \in D} \Theta_d$$

Suppose that for each $d \in D$

$$\{X : T_d(X) \in R_d\}$$

is rejection region for a test of $H_{0d} : \theta \in \Theta_d$ vs $H_{1d} : \theta \in \Theta_d^c$. Then the rejection region for IUT of H_0 vs H_1 is

$$\bigcap_{d \in D} \{X : T_d(X) \in R_d\}$$

So H_0 can be rejected $\Leftrightarrow$ Each H_{0d} can rejected

Remark: If the rejection region are all of the form

$$\{X : T_d(X) \geq c\} \quad (c \text{ indep of } d)$$

the rejection region for H_0 is

$$\bigcap_{d \in D} \{X : T_d(X) \geq c\} = \{X : \inf_{d \in D} T_d(X) \geq c\}$$

$$\begin{tikzcd} A \arrow[r, "\alpha_1"'] \arrow[rr, bend left, "\theta_2"] & M \arrow[r, "\theta_1"'] & Y \end{tikzcd}$$

$$E(Y \mid M, A) = \theta_0 + \theta_1 M_1 + \theta_2 A$$

$$E(M \mid A) = \alpha_0 + \alpha_1 A$$

H_0 : Indirect Effect $= 0 \Leftrightarrow H_0 : \theta_1 \alpha_1 = 0$

Double A

Subject: ...... Date: ..../..../....

We define $H_0^{(1)}: \alpha_1 = 0,\ \theta_1 \neq 0$

$H_0^{(2)}: \alpha_1 \neq 0,\ \theta_1 = 0$

$H_0^{(3)}: \alpha_1 = 0,\ \theta_1 = 0$

and the test of indirect effects is conducted under a composite null hypothesis

$$H_0 = H_0^{(1)} \cup H_0^{(2)} \cup H_0^{(3)}$$

against

$$H_1 = \alpha_1 \neq 0,\ \theta_1 \neq 0$$

☆ Product significant test (Sobel Test)

$$U_{pt} = \frac{[\hat{\alpha}_1 \hat{\theta}_1]^2}{\hat{\alpha}_1^2 V(\hat{\theta}_1) + \hat{\theta}_1^2 V(\hat{\alpha}_1)} \overset{H_0}{\sim} \chi^2_{(1)}$$

☆ Joint significant test (IUT)

$$U_{jt} = \frac{(\hat{\alpha}_1)^2}{V(\hat{\alpha}_1)} \wedge \frac{(\hat{\theta}_1)^2}{V(\hat{\theta}_1)} \overset{H_0}{\sim} \chi^2_{(1)}$$

(Remark. $U_{jt} \geq U_{pt}$ since $\frac{|ab|}{\sqrt{a^2+b^2}} \leq |a| \wedge |b|$

$P_{jt} \leq P_{pt}$)

However, joint significant tests still have a considerably conservative departure from the theoretical null.

Ex.

$$\underset{A}{\text{Stress}} \longrightarrow \underset{M}{\text{Sleep Quality}} \longrightarrow \underset{Y}{\text{Job performance}}$$

lm(M ~ A) lm(Y ~ M+A)

Both $p < 0.05 \Rightarrow$ Mediation

Double A

§ Separable Effect

Exposure A decomposed into two separable components (S^Y, S^M)

$$\begin{tikzcd} A \arrow[r] \arrow[dr] & S^M \arrow[r] \arrow[dr, dashed] & M \arrow[d] \\ & S^Y \arrow[r] \arrow[ur, dashed] & Y \end{tikzcd}$$

(Dashed edge signify the presence of ind-level effects but absent pop-level)

(S^Y, S^M) are not observed, and observe only the value of A.

Determinism

An individual is considered (un)exposed to A

$$A = a^*$$
$$\Updownarrow$$
$$S^Y = a^*,\ S^M = a^*$$

This individual is (un)exposed to all components simultaneously

Compared to previous defined counterfactuals

$$Y(a,m),\ M(a)$$

Similar we can define

$$Y_S(s^Y, s^M, m),\ M_S(s^Y, s^M)$$

(Non-cross world counterfactuals)

Based on determinism

$$Y_S(a^*, a^*, m) = Y(a^*, m)$$
$$M_S(a^*, a^*) = M(a^*)$$

Assume the counterfactuals under separable effect follow the composition assumptions

$$Y_S(S^Y, S^M, M_S(S^Y, S^M)) = Y(S^Y, S^M)$$

The total effect (TE) can be decomposed into separable direct effect (SDE) and separable indirect effect (SIE)

$$\phi_S(a, a^*) = \mathbb{E}[Y_S(a, a^*)]$$

$$\begin{aligned} TE &= E[Y_S(1,1) - Y_S(0,0)] \\ &= E[(Y_S(1,1) - Y_S(1,0)) \\ &\qquad + (Y_S(1,0) - Y_S(0,0))] \\ &= SIE + SDE \end{aligned}$$

SDE: effect of S^Y component on Y

SIE: effect of S^M component on Y

★ Dismissible Component assumption

DC-1 : $E[Y_S(a,a^*) \mid M_S(a,a^*) = m] = E[Y_S(a,a) \mid M_S(a,a) = m]$
(No DE of S^M on Y)

DC-2 : $P(M_S(a,a^*) = m) = P(M_S(a^*,a^*) = m)$
(No DE of S^Y on Y)

Thm. Under DCs assumption and $A \perp\!\!\!\perp (M(a), Y(a))$
the $\phi_S(a,a^*)$ is identified as mediation formula

★ Identification of $\phi_S(a,a^*)$

$$\begin{aligned} E[Y_S(a,a^*)] &= \sum_m E[Y_S(a,a^*) \mid M_S(a,a^*) = m] \, P(M_S(a,a^*) = m) \\ &\overset{\text{DCs}}{=} \sum_m E[Y_S(a,a) \mid M_S(a,a) = m] \, P(M_S(a^*,a^*) = m) \\ &\overset{\text{determinism}}{=} \sum_m E[Y(a) \mid M(a) = m] \, P(M(a^*) = m) \\ &\overset{A \perp\!\!\!\perp (M(a), Y)}{=} \sum_m E[Y(a) \mid M(a) = m, A = a] \, P(M(a^*) = m \mid A = a^*) \\ &\overset{\text{consistency}}{=} \sum_m E[Y \mid M = m, A = a] \, P(M = m \mid A = a^*) \end{aligned}$$

DCs offer several advantages :
they are verifiable, imply pop-level isolation assumptions and allow identifiability of separable effects when $A \perp\!\!\!\perp (M(a), Y(a))$

ξ Controlled Direct Effect (CDE)

CDE compares the potential outcomes to assigning treatment $A=a$ w/ that of assigning treatment $A=a^*$ while controlling the mediator at $M=m$

$$CDE(m) = E[Y(a,m)] - E[Y(a^*,m)]$$

"m fixed", $A \uparrow_{a}^{a^*}$

★ It is often not realistic to image scenarios where one would consider forcing the mediator to be same for all subjects in the population.

Ind 1 $\begin{cases} CDE(0) & Y_1(1,0) - Y_1(0,0) \\ \neq & \\ NDE & Y_1(1,M_1(0)) - Y_1(0,M_1(0)) \end{cases}$ if $M_1(0)=1$

Ind 2 $\begin{cases} CDE(0) & Y_2(1,0) - Y_2(0,0) \\ \parallel & \\ NDE & Y_2(1,M_2(0)) - Y_2(0,M_2(0)) \end{cases}$ if $M_2(0)=0$

★ IE can't to be defined in a similar manner as CDEs because it is impossible to hold a set of variables fixed.

★ $\left(E[Y(1)] - E[Y(0)]\right) - \left(E[Y(1,m)] - E[Y(0,m)]\right)$ may not carry the interpretation of IE.

$\Updownarrow$

Because of potential exposure-mediator interactions, the difference may be nonzero even if the exposure has no effect on the mediator

$CDE \overset{?}{\longleftrightarrow} CIE$ (x)

★A proportion measure that would be of policy relevance ↑ Proportion mediated

$$PE(m) = \frac{TE - CDE(m)}{TE} \quad \left(\Leftrightarrow \ PM = \frac{NIE}{TE} \times 100\% \right)$$

proportion of effect of the exposure on the outcome that could be eliminated by intervening to set the intermediate to some fixed level m.

$$\left(\underset{\text{Not Control M}}{\underset{\downarrow}{TE}} - \underset{\text{Control M}}{\underset{\downarrow}{CDE(m)}} \right),$$

(If PE were large and we want to prevent the effect of the exposure on the outcome, we might try to implement policies to intervene on the intermediate.)

★ Identification of CDE

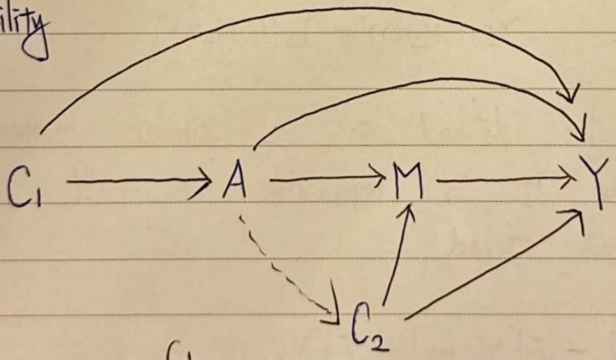
$$\phi(a,m) = \mathbb{E}[Y(a,m)]$$

$$CDE(m) = \phi(1,m) - \phi(0,m)$$

1. Consistency

$$Y(a,m) = Y \quad \text{if} \quad \begin{cases} A = a \\ M = m \end{cases}$$

2. Exchangeability



i) $A \perp\!\!\!\perp Y(a,m) \mid \overset{C_1}{C}$ No unmeasured A-Y confounding

ii) $M \perp\!\!\!\perp Y(a,m) \mid \underset{C_2}{C}, A$ No unmeasured M-Y confounding

$$\mathbb{E}[Y(a,m)] = \sum_c \mathbb{E}[Y(a,m) \mid C=c] \, P(C=c)$$

$$\left(A \perp\!\!\!\perp Y(a,m) \mid C \right) = \sum_c \mathbb{E}[Y(a,m) \mid A=a, C=c]$$

$$\left(M \perp\!\!\!\perp Y(a,m) \mid A, C\right) = \sum_c E[Y(a,m) \mid M=m, A=a, C=c]\, P(C=c)$$

$$(\text{Consistency}) \quad = \sum_c E[Y \mid M=m, A=a, C=c]\, P(C=c)$$

$$= Q(a,m)$$

$$\phi(a,m) \triangleq Q(a,m)$$

Consider $E[Y \mid M=m, A=a, C=c] = \theta_0 + \theta_1 a + \theta_2 m + \theta_c C$

$$CDE(m) = \theta_1 (a - a^*)$$

§ Binary Outcome

A, Y is Binary

Risk $\begin{cases} P(Y=1 \mid A=1) = R_1 \\ P(Y=1 \mid A=0) = R_0 \end{cases}$

i) Risk Difference (RD)

$$R_1 - R_0$$

ii) Risk Ratio (RR)

$$R_1 / R_0$$

iii) Odds Ratio (OR)

$$O_1 / O_0 \qquad O_i = \frac{R_i}{1-R_i}$$

For i), $E(Y|A) = \theta_0 + \theta_1 A$

$$\left.\begin{aligned} R_1 &= \theta_0 + \theta_1 \\ R_0 &= \theta_0 \end{aligned}\right\} \Rightarrow RD = \theta_1$$

ii), $\log[E(Y|A)] = \theta_0 + \theta_1 A$

$$\left.\begin{aligned} \log(R_1) &= \theta_0 + \theta_1 \\ \log(R_0) &= \theta_0 \end{aligned}\right\} \Rightarrow RR = e^{\theta_1}$$

iii), $\operatorname{logit}[E(Y|A)] = \theta_0 + \theta_1 A$

$$\left.\begin{aligned} \operatorname{logit}(R_1) &= \theta_0 + \theta_1 \\ \operatorname{logit}(R_0) &= \theta_0 \end{aligned}\right\} \Rightarrow OR = e^{\theta_1}$$

when outcome is rare (Rule of thumb: $\phi \le 0.1$)

$$\frac{\phi}{1-\phi} \approx \phi, \quad \text{Odds} \approx \text{Risk}$$

$P(Y=1|A=1)$ vs $P(Y=1|A=0)$ Association

individual $Y_i(1)$ vs $Y_i(0)$ counterfactual

population $E[Y(1)]$ vs $E[Y(0)]$ causality

$$E[Y(1)] \Downarrow P(Y(1)=1) \;\approx\; R(1) \qquad\qquad E[Y(0)] \Downarrow P(Y(0)=1) \;\approx\; R(0)$$

id	A	Y	Y(1)	Y(0)
1	1	1		
2	1	0		
3	1	1		
4	0	0		
5	0	1		
6	0	0		

Y (A = 1) → R(1)

Y (A = 0) → R(0)

$\phi(a) = E[Y(a)] = R(a)$

$\Rightarrow$ counterfactual

$$RD = R(1) - R(0)$$

$$RR = R(1) / R(0)$$

$$OR = O(1) / O(0), \quad O(i) = \frac{R(i)}{1-R(i)}$$

★ if $\phi(a,a^*) = E[Y(a, M(a^*))]$

$$TE_{RD} = \phi(1,1) - \phi(0,0) = [\phi(1,1) - \phi(1,0)] - [\phi(1,0) - \phi(0,0)]$$

$$= NIE_{RD} + NDE_{RD}$$

$$TE_{RR} = \frac{\phi(1,1)}{\phi(0,0)} = \frac{\phi(1,1)}{\phi(1,0)} \times \frac{\phi(1,0)}{\phi(0,0)} = NIE_{RR} \times NDE_{RR}$$

$$TE_{OR} = \left[\frac{\phi(1,1)}{1-\phi(1,1)}\right] / \left[\frac{\phi(0,0)}{1-\phi(0,0)}\right] = \frac{O(1,1)}{O(1,0)} \times \frac{O(1,0)}{O(0,0)}$$

$$= NIE_{OR} + NDE_{OR}$$

In toxicology, binary response models describe the effect of dosage of a toxin on whether a subject die.

Let τ denote the dosage level. For a subject i, a tolerance threshold T_i is assume related to response y_i by

$$y_i = \begin{cases} 1 & \text{if } T_i > \tau \\ 0 & \leq \tau \end{cases} \quad \text{(subject die if dosage exceed } \tau\text{)}$$

assume that

$$T_i = \underset{\sim}{x_i}^T \underset{\sim}{\beta} + \varepsilon_i$$

where $\underset{\sim}{x_i}^T = (1, x_{1i}, \ldots, x_{pi})$

and $E(\varepsilon_i) = 0$, ε_i has cdf F

Thus,

$$\begin{aligned} P(Y_i = 1) &= P(T_i > \tau) \\ &= P(\underset{\sim}{x_i}^T \underset{\sim}{\beta} + \varepsilon_i > \tau) \\ &= P(\varepsilon_i > \tau - \underset{\sim}{x_i}^T \underset{\sim}{\beta}) \\ &= 1 - P(\varepsilon_i \leq \tau - \underset{\sim}{x_i}^T \underset{\sim}{\beta}) \\ &= 1 - F(\tau - \underset{\sim}{x_i}^T \underset{\sim}{\beta}) \end{aligned}$$

when $\tau = 0$ and F symmetric at zero (Even fn)

$$P(Y_i = 1) = F(\underset{\sim}{x_i}^T \underset{\sim}{\beta})$$

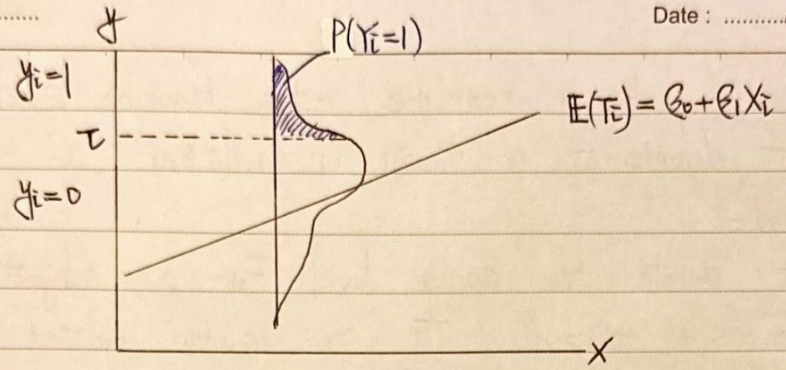
$$\Updownarrow$$

$$F^{-1}(P(Y_i = 1)) = \underset{\sim}{x_i}^T \underset{\sim}{\beta}$$

When F is strictly increasing over $\mathbb{R}$, its inverse F^{-1} exist. F maps $(0,1)$ range of prob'y onto $\mathbb{R}$, the range of linear preditors.

F is cdf of $N(0,1) \Rightarrow$ Probit model

" logistic dist^n $\Rightarrow$ Logistic model



The logit and probit link are symmetric at 0.5, in the sense that

$$F(p) = -F(1-p)$$

$$\left(\text{logit}(p) = \log\left(\frac{p}{1-p}\right) = -\log\left(\frac{1-p}{p}\right) = -\text{logit}(1-p) \right)$$

These models are inappropriate when this is badly violated.

A different shape of response curve is

$$\log(-\log(1-p_i)) = \underset{\sim}{x_i}^T \underset{\sim}{\beta}$$

$$\left(\begin{array}{c} 1-p_i \in (0,1) \\ \log(1-p_i) \in (-\infty, 0) \\ -\log(1-p_i) \in (0,\infty) \\ \log(-\log(1-p_i)) \in \mathbb{R} \end{array} \right)$$

It is asymmetric,
$\Rightarrow$ p_i approaching 0 fairly slowly
 〃 1 quite sharply

The link fn is called Complementary log-log. (Cloglog)

★ Identity link $F(p)=p$ is sometimes used for binomial data to yield a linear prob'y model.
However, the identity link can predict nonsense probabilities.
$\hat{p} \notin (0,1)$

★ GLM (Generalized Linear Model)

a. Random Component $f(\underline{y})$

b. Linear Predictor $X\underset{\sim}{\beta}$

c. Link function g such that $g(E(Y)) = X\underset{\sim}{\beta}$

★ Exponential Dispersion Family

natural parameter

$$f(y \mid \theta, \phi) = \exp\left\{ \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right\}$$

dispersion parameter

where $\mu = E(Y) = b'(\theta)$

$Var(Y) = b''(\theta) a(\phi)$

Ex: $Y \sim Poi(\lambda)$

$$f(y \mid \lambda) = \frac{e^{-\lambda}\lambda^y}{y!} = e^{y\ln\lambda - \lambda - \ln(y!)}$$

$$\theta = \ln\lambda \Rightarrow \lambda = e^{\theta} = b(\theta)$$

$$a(\phi) = 1, \quad c(y, \phi) = -\ln(y!)$$

$$\mu = E(Y) = b'(\theta) = e^{\theta} \Rightarrow \log \text{ link}$$

Remark. $Y \sim$ Binomial $\Rightarrow$ logit link

$Y \sim$ Normal $\Rightarrow$ identity link

GLM states that linear predictor relates to μ by $\theta = g(\mu)$ for a link function g.

Canonical Link $\Leftrightarrow$

Probit

$F^{-1}(p) = X\underset{\sim}{\beta}$, F is cdf of $N(0,1)$

Complementary Log-Log

$\log(-\log(p)) = X\underset{\sim}{\beta}$